STRATEGY

07

AI Chatbot Strategy

Business Purpose, Use Cases & Implementation Plan

PREPARED FOR	PREPARED BY	DATE	DOCUMENT
Precision Structures Inc.	EBYG Automation — SSC Digital Services	March 19, 2026	07 of 17

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1. Business Purpose

Precision Structures receives phone calls and emails for questions that often have straightforward answers: service area, hours, how the bid process works, what types of trusses are available. An AI chatbot on the website serves as a first point of contact that can:

- **Qualify leads 24/7** — capture project details from visitors outside business hours (Mon-Fri 8am-4pm)
- **Answer common questions instantly** — reduce the volume of phone calls for simple inquiries
- **Guide the bid process** — explain what information is needed and how to submit a request
- **Build trust before first contact** — surface the company's strengths (family-owned since 1990, 35+ years experience, 6-state service area) naturally in conversation
- **Route inquiries correctly** — send bid requests to bids@precisionstructures.net, general questions to contact@precisionstructures.net, and urgent matters to the phone line

The chatbot is not a replacement for human interaction. It is a front door that makes the company more accessible and ensures no inquiry falls through the cracks.

2. What the Chatbot Should Do

- Answer questions about services (roof trusses, floor trusses, wall panels, engineered wood products, truss repair/reinforcement)
- Explain truss terminology in plain language when talking to homeowners
- Walk visitors through the bid submission process step by step
- Collect basic project information: type, location, timeline, contact details
- Provide business hours, address, phone number, and email
- Clarify the service area (Utah, Idaho, Wyoming, Colorado, Nevada, Montana)
- Explain what makes Precision Structures different (experience, family-owned, quality focus)
- Offer to connect the visitor with a human when the question goes beyond chatbot scope
- Share general process information: how long does a typical project take, what to expect after submitting plans

3. What the Chatbot Should NOT Do

- **Provide binding quotes or specific pricing.** Truss pricing depends on engineering, materials, and project

- **Give engineering advice.** Never recommend specific truss configurations, load ratings, span capabilities, or structural solutions.
- **Make promises about timelines.** Timelines depend on workload, complexity, and material availability. The chatbot can share general ranges but must frame them as estimates.
- **Discuss competitors.** No comparisons, no opinions on other companies, no commentary on the market.
- **Handle complaints or disputes.** Escalate immediately to a human.
- **Discuss pricing strategies, margins, or internal operations.**
- **Provide legal, insurance, or warranty specifics** beyond directing to the appropriate contact.

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4. Top Use Cases with Example Flows

Use Case 1: Homeowner Planning a Build

> ***Visitor:*** *"I'm building a house and need roof trusses. How does this work?"* ***Bot:*** *Explains the general process: submit plans or rough dimensions, receive a bid, engineering and fabrication, delivery. Asks if they have plans yet. Offers to explain what info is needed for a bid.*

Use Case 2: Contractor Requesting a Bid

> ***Visitor:*** *"I need a bid for a 4,200 sq ft custom home, plans ready."* ***Bot:*** *Collects project type, location, timeline. Provides bids@precisionstructures.net for plan submission. Explains typical turnaround for bids. Asks if they need floor trusses and wall panels too.*

Use Case 3: After-Hours Inquiry

> ***Visitor (Saturday 9pm):*** *"Are you guys open?"* ***Bot:*** *Shares hours (Mon-Fri 8am-4pm). Offers to collect their question/project info so the team can follow up Monday morning. Captures name, phone/email, and brief description.*

Use Case 4: Terminology Question

> ***Visitor:*** *"What's the difference between a truss and a rafter?"* ***Bot:*** *Explains clearly: a truss is a pre-engineered, factory-built structural frame; a rafter is a single beam cut and installed on-site. Trusses are typically faster to install and engineered for specific loads. Offers more detail if asked.*

Use Case 5: Service Area Check

> ***Visitor:*** *"Do you deliver to Boise?"* ***Bot:*** *Confirms Idaho is in the service area. Suggests submitting a bid request for a delivery estimate specific to their location.*

Use Case 6: Employment Inquiry

> ***Visitor:*** *"Are you hiring?"* ***Bot:*** *Directs to contact@precisionstructures.net or suggests calling (801) 985-3000 during business hours. Mentions this is a family-owned company that values quality and craftsmanship.*

Use Case 7: Urgent/Complex Engineering Question

> ***Visitor:*** *"Can your trusses span 60 feet clear? I need a specific load rating for my commercial project."* ***Bot:*** *Acknowledges the question, explains that engineering specifics require a direct conversation with the team, and provides phone number and email. Does not guess at capabilities.*

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5. Content Sources

Already in the System

- Company overview (history, family-owned, years in business)
- Service descriptions (roof trusses, floor trusses, wall panels, etc.)
- Contact information (phone, email, address, hours)
- Service area (6 states)

Should Be Added

- **FAQ content:** 15-20 most common questions and approved answers
- **Terminology glossary:** Truss types (Fink, Howe, scissor, mono, etc.), chord, web, span, pitch, bearing point, heel height, overhang
- **Bid process details:** What to submit, what to expect, typical timelines
- **Project type guidance:** Residential vs. commercial considerations
- **Seasonal information:** Busy seasons, lead time expectations

Content Maintenance

- Review and update chatbot knowledge base quarterly
- Add new FAQs based on common chatbot escalations
- Update seasonal messaging as needed (e.g., winter lead times)

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6. Guardrails

Response Length

- Default: 2-3 sentences per response
- Longer explanations (terminology, process walkthroughs): up to 5-6 sentences, broken into digestible pieces
- Never send a single response longer than ~150 words without breaking it up or offering to continue

Stay On Topic

- The chatbot discusses Precision Structures services, trusses, the construction industry as it relates to the company's offerings, and general building process information
- Off-topic questions receive a polite redirect: "I'm best equipped to help with questions about trusses and our services. For other questions, you can reach our team at (801) 985-3000."

No Fabrication

- If the chatbot does not have a confident answer, it says so and routes to a human
- Never invent pricing, timelines, specifications, or capabilities
- When uncertain, default to: "For the most accurate answer on that, I'd recommend contacting our team directly."

Tone Consistency

- Maintain professional, approachable tone across all interactions
- No sarcasm, no humor that could be misread, no excessive enthusiasm
- Match the visitor's energy level — if they're brief, be brief; if they're detailed, provide more detail

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7. Escalation Triggers

The chatbot should hand off to a human (or strongly recommend calling/emailing) when it detects:

TRIGGER	RESPONSE
Angry or frustrated customer	Empathize, apologize for any inconvenience, provide phone number for immediate help
Complex engineering question	Acknowledge the complexity, explain this needs the engineering team, provide contact info
Specific pricing request	Explain that pricing depends on project specifics, guide toward bid submission
Complaint about work or service	Express concern, provide phone number and contact@precisionstructures.net, do not attempt to resolve
Legal question	Do not engage; direct to contact@precisionstructures.net
Safety concern	Take seriously, direct to phone immediately
Repeated questions not being answered	Recognize the loop, offer to connect with a human
Request to speak with a person	Provide phone number and hours immediately, no resistance

Escalation language: "I want to make sure you get the best help on this. Let me connect you with our team — you can reach them at (801) 985-3000 during business hours (Mon-Fri, 8am-4pm) or email contact@precisionstructures.net anytime."

8. Legal and Safety Considerations

Required Disclaimer

The chatbot must identify itself as AI in its greeting. Example: "Hi, I'm the Precision Structures assistant — an AI helper for quick questions about our services."

Engineering Disclaimer

Any discussion of structural capabilities, load ratings, or engineering topics must include: "This is general information only. All structural decisions should be made by a licensed engineer. Our team can connect you with engineering support for your specific project."

Non-Binding Estimates

Any mention of timelines, general cost ranges, or project duration must include language indicating these are rough estimates and not commitments: "Timelines vary based on project complexity and current workload. For a specific estimate, please submit your plans to our bid team."

Data Collection

- Inform visitors that any information they share (name, email, phone, project details) will be used to follow up on

- Do not request or store sensitive information (SSN, financial data, etc.)
- Comply with applicable privacy regulations

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9. Current State Assessment

What Exists

- `chatbot.js` — 321 lines of client-side JavaScript
- Claude API integration with fallback keyword matching
- Basic UI with chat bubble and conversation window
- System prompt with company information
- Keyword-based responses for common topics (hours, contact, services, etc.)

What Works Well

- Fallback keyword matching ensures basic questions get answers even without API
- Company information is embedded in the system prompt
- UI is functional and matches the site design

What Needs Improvement

- **Backend proxy required:** The Claude API should not be called directly from client-side JavaScript. API keys in the browser are a security risk. An Azure Function proxy is needed.
- **Persona refinement:** The current system prompt needs the persona definition (see companion document: `precision-chatbot-persona.md`)
- **Response quality:** With a refined system prompt and better guardrails, response quality will improve significantly
- **Lead capture:** Currently no mechanism to collect and store visitor information for follow-up
- **Analytics:** No tracking of chatbot usage, common questions, or escalation patterns
- **Mobile experience:** Verify chatbot works well on mobile devices (common for contractors on job sites)

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10. Implementation Phases

Phase 1: Launch (Weeks 1-4)

Goal: Secure, reliable chatbot with refined persona and proper architecture.

TASK	DETAILS
Backend proxy	Azure Function to proxy Claude API calls; remove API key from client-side code
Persona update	Implement the persona definition from `precision-chatbot-persona.md`
System prompt refinement	Incorporate FAQ content, guardrails, escalation rules
Fallback improvements	Expand keyword matching for cases when API is unavailable
Testing	Test against top 20 most common questions; verify escalation triggers work
Mobile QA	Verify chat experience on iOS and Android browsers
Disclaimer	Add AI identification to greeting

Success criteria:

- API key not exposed in client-side code
- Chatbot correctly answers 90%+ of common questions
- Escalation triggers fire correctly
- Response time < 3 seconds

Phase 2: Enhanced (Weeks 5-12)

Goal: Lead capture, analytics, and continuous improvement.

TASK	DETAILS
Lead capture form	Collect name, email/phone, project type, timeline within chat flow
Lead storage	Azure Table Storage or simple database for captured leads
Email notifications	Auto-email the team when a lead is captured or escalation triggered
Analytics dashboard	Track: conversations/day, top questions, escalation rate, lead capture rate
CRM integration	If applicable, push captured leads to existing CRM system
Conversation history	Store anonymized conversations for quality review and prompt improvement
A/B testing	Test different greetings, response styles, and lead capture prompts
Knowledge base expansion	Add terminology glossary, seasonal content, expanded FAQ

Success criteria:

- Lead capture rate > 15% of conversations
- Average conversation length > 3 messages (indicates engagement)
- Escalation rate < 10% of conversations
- Team receives lead notifications within 5 minutes

11. Cost Estimate

Phase 1

- Azure Function: Free tier covers expected volume (up to 1M executions/month)
- Claude API: ~$20-50/month estimated based on moderate chat volume
- Development time: 15-25 hours

Phase 2

- Azure Table Storage: < $5/month
- Additional Azure Function for lead processing: covered by free tier
- Analytics tooling: depends on approach (basic custom dashboard vs. third-party)
- Development time: 30-50 hours

Ongoing

- Claude API usage: scales with traffic, monitor monthly
- Content updates: 2-4 hours/quarter for FAQ and knowledge base maintenance
- Prompt tuning: periodic adjustments based on conversation review

12. Risks and Mitigations

RISK	LIKELIHOOD	IMPACT	MITIGATION
API key exposed in client code	High (current state)	High	Phase 1 priority: backend proxy
Chatbot gives incorrect information	Medium	High	Guardrails, human review of conversations, disclaimer
Low adoption / visitors ignore chatbot	Medium	Low	Test positioning, greeting variations, mobile experience
High API costs from abuse	Low	Medium	Rate limiting on proxy, conversation length limits
Chatbot creates negative impression	Low	High	Professional persona, thorough testing, easy escalation to human